Aarush Ram Anandh

aarush.anandh_ug2024@ashoka.edu.in | +91 9659786783
github.com/MushuRam01 | linkedin.com/in/aarush-ram-anandh

11 November 2026

Hiring Team
IBM Research / Software Innovation Lab India

Application: AI Research / AI & Data Internship

Dear Hiring Team,

I am applying for the AI research and AI & Data tracks of IBM's 2027 summer internship program in India. I am completing a B.Sc. (Hons.) in Computer Science and Mathematics at Ashoka University, expected in 2028. My experience is in generative models, reinforcement learning and time-series modelling, and I am particularly interested in research that combines controlled experiments with usable implementations.

At the Constrained Image-Synthesis Lab, I improved colour fidelity in latent diffusion models to $\Delta E_{00} < 1$, with approximately four times better colour accuracy than evaluated open-source baselines. This work is reported in the public preprint *Inference-Time Loss-Guided Colour Preservation in Diffusion Sampling* (arXiv:2601.17259). I am also developing Controlled Generalised Sliced Wasserstein, a controllable optimal-transport framework with a paper submitted to JMLR; my work includes computational methods in C++ and specifications and proofs in Lean.

In a two-person RL coursework project, we co-developed a Gymnasium/poke-env environment with legal-action masks, potential-based reward shaping, MaskablePPO and masked DQN agents. We evaluated PPO over two variants and five seeds each, reaching 99.7–100% wins against the fixed heuristic over 1,020 final battles per run. In a separate team project with Tanush Seal, I worked on financial forecasting using a CatBoost baseline and HMM, Transformer and other residual models, with company-level diagnostic dashboards. At the Koita Centre for Digital Health, I contributed to a wearable-data pipeline achieving approximately 70% sensitivity for asthma exacerbation detection in study evaluation, and built experimental frozen time-series foundation-model routes for surprise features and embeddings. Those foundation-model routes were implemented but not yet run in the reported cohort evaluation.

At Avartan Labs, I raised rare-class correct detection from 40% to 80% in an RT-DETR pipeline covering 18 material classes. This industry internship gave me experience analysing detector failures and investigating targeted model improvements.

The emphasis on model behaviour, reinforcement learning, data and experimental research in IBM's AI tracks fits the direction I want to pursue. I would welcome the opportunity to contribute Python/PyTorch implementation and evaluation work, while learning how these methods support IBM's research and enterprise applications. Thank you for considering my application.

Yours sincerely,

Aarush Ram Anandh